

Brain tumour classification from MRI using ResNet18

A two-phase transfer-learning schedule — frozen backbone, then a fine-tuned deepest block — takes a ResNet18 to 97.00% accuracy on four-way brain MRI classification from only 5,000 training images.

97.00%

Test accuracy

0.9715

Macro F1

100%

No-tumour recall

30

Errors in 1,000

1. PROBLEM

Tumour type must be identified before treatment can be planned. Manual MRI reading is slow, demands scarce expertise, and shows measurable inter-observer variability — particularly between **glioma** and **meningioma**, which can present with overlapping appearance on a single axial slice.

The task here is single-label, four-way classification of one axial slice into glioma, meningioma, no tumour, or pituitary tumour.

2. DATA

SPLIT	IMAGES	DERIVATION
Train	4,000	80% of train dir
Validation	1,000	20%, stratified
Test	1,000	held out entirely

The 80:20 split is stratified on the label so class balance is preserved in both halves, and seeded (`random_state = 42`) so the run reproduces exactly. The classes are moderately imbalanced — no tumour is only 14% of the test set — which is why §5 reports per-class metrics rather than accuracy alone.

3. PREPROCESSING

STEP	OPERATION	WHY
1	Resize 224×224	ResNet18's pretrained resolution
2	Greyscale → 3ch	Pretrained RGB filters apply unchanged
3	Horizontal flip	Train only; anatomy is near-symmetric
4	ImageNet normalise	Matches the pretrained distribution

On augmentation. Only horizontal flipping is used. Rotation, shear and colour jitter were deliberately excluded: they can introduce geometry or intensity patterns that never occur in real acquisition, and an augmentation that fabricates non-physical images teaches the model artefacts rather than pathology. Validation and test transforms use no augmentation at all.

4. METHOD

ResNet18 pretrained on ImageNet, with the final layer replaced: `Linear(512 → 1000)` becomes `Linear(512 → 4)`. Total parameters: 11,178,564.

That new head starts **random**. Unfreezing the whole network immediately would push large, noisy gradients back through the backbone and corrupt the pretrained features before they could be of any use. The two-phase schedule prevents this.

SETTING	PHASE 1	PHASE 2
Epochs	5	15
Trainable params	2,052	8,395,780
Learning rate	1e-4	5e-5
Scheduler	none	StepLR ($\gamma=0.1$, step 7)
Unfrozen	fc only	layer4 + fc
Optimiser	Adam	Adam
Batch size	32	32

Only **layer4** is unfrozen in Phase 2, not the full backbone: early convolutional layers encode generic edge and texture detectors that transfer well across domains, while the deepest block encodes the class-specific semantics that most need adapting from natural images to MRI.

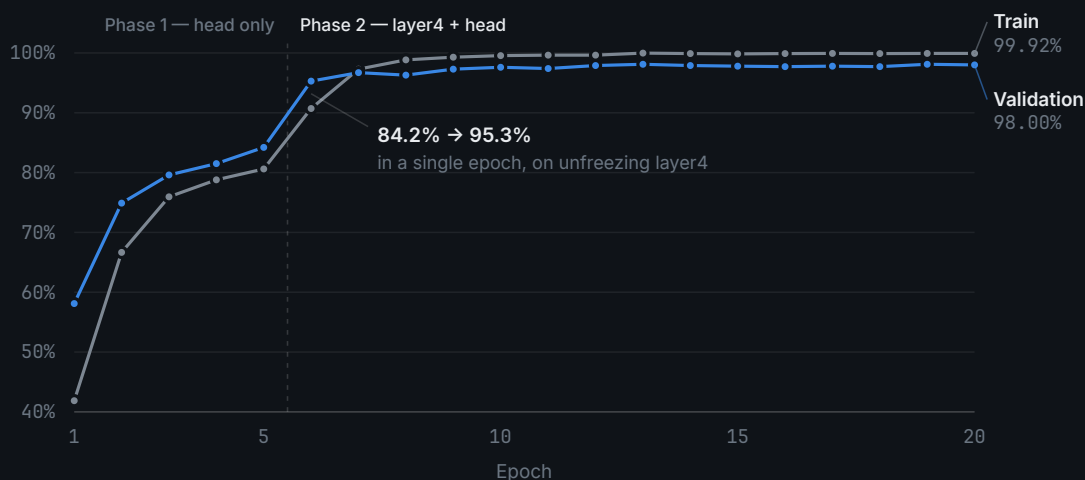


Figure 1 — Accuracy across both phases. Phase 1 plateaus at 84.2% validation accuracy; a frozen ImageNet backbone can only reach so far into a domain as distant as MRI. Unfreezing layer4 lifts validation accuracy to 95.3% within a single epoch and past 97% by epoch 9. From epoch 13 onward, training sits at 99.9%+ while validation holds a narrow 97.7–98.1% band and validation loss stops falling — the model has converged, and further epochs would not help.

Epoch-by-epoch

5. RESULTS

Test accuracy: 97.00% — 970 of 1,000 held-out scans classified correctly.

CLASS	PRECISION	RECALL	F1	SUPPORT
■ Glioma	0.9793	0.9331	0.9556	254
■ Meningioma	0.9486	0.9641	0.9562	306
■ No tumour	0.9722	1.0000	0.9859	140
■ Pituitary	0.9835	0.9933	0.9884	300
Macro avg	0.9709	0.9726	0.9715	1,000
Weighted avg	0.9702	0.9700	0.9699	1,000

Macro and weighted averages agree to within 0.002, which confirms the headline number is not being carried by the larger classes — the model is genuinely competent on all four.

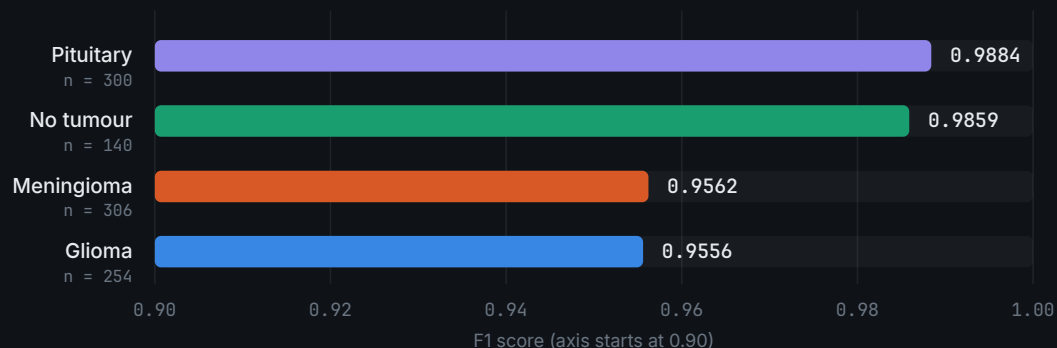


Figure 2 — Per-class F1. Pituitary and no tumour lead; glioma and meningioma trail together, which is the first hint that they are being confused with each other rather than failing independently.

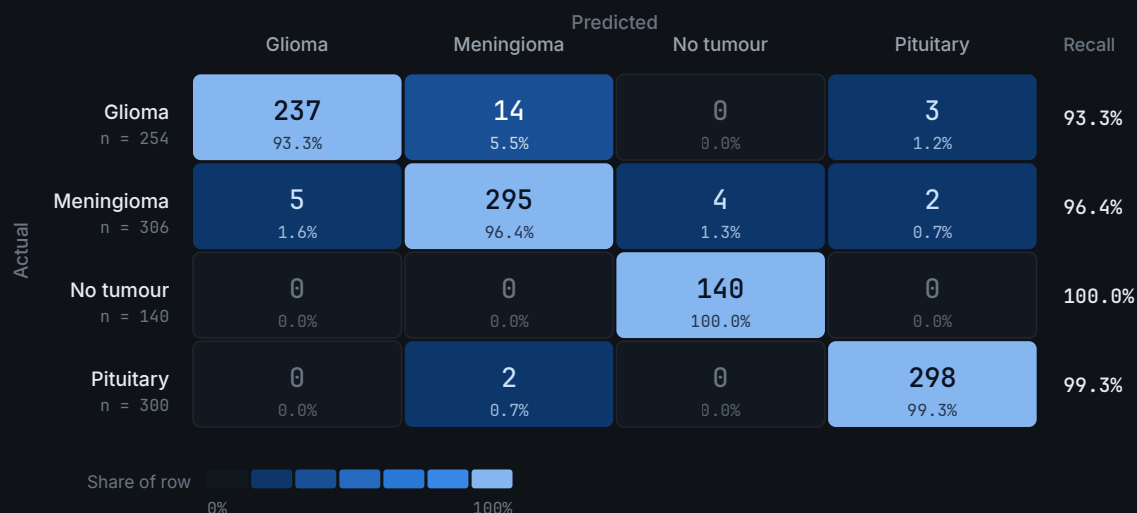


Figure 3 — Confusion matrix. Rows are the true class, columns the prediction; cells carry the raw count and the share of the row, and the diagonal is per-class recall. Thirty errors in total, and their distribution is anything but uniform.

Error analysis

- **Glioma ↔ meningioma accounts for 19 of 30 errors (63%).** Fourteen gliomas read as meningioma, five meningiomas as glioma. This is the single dominant failure mode, and it mirrors the hardest distinction for human readers: on one axial slice, without contrast timing or multi-plane context, the two can look genuinely similar. Glioma consequently has the lowest recall of the four, at 93.3%.
- **No tumour achieves perfect recall (140/140).** Not one scan containing a tumour was classified as tumour-free. For a screening-style application this is the most valuable property the model has, since a false negative is the costliest error. Its precision of 0.9722 reflects four meningiomas misfiled as no-tumour — errors in the *other* direction, which are the safer kind.
- **Pituitary is nearly separable**, at 99.3% recall with two errors in 300. Pituitary tumours occupy a distinctive skull-base location, giving the network a positional cue the other tumour types do not offer.

6. DEPLOYMENT

The trained weights are served by a Flask application. Device selection is automatic at start-up — CUDA on an NVIDIA GPU, Apple MPS on Apple Silicon, otherwise CPU. The original training run used Apple MPS.

METHOD	ROUTE	PURPOSE
GET	/	Browser interface
GET	/health	Liveness + active device
POST	/predict	multipart image → prediction

The uploaded image runs through the same `resize → greyscale-to-3-channel → normalise transform` used at validation time, then through the network under `torch.no_grad()`, softmaxed into a distribution.

The interface renders the **full probability distribution**, not just the winning class. A 97%-confident prediction and a 51%-confident one are very different objects, and surfacing the distribution makes that difference visible to whoever reads the result. Class colours are fixed and contrast-validated, and every bar is directly labelled, so identity never rests on colour alone.

7. LIMITATIONS

- **Single-slice classification.** Radiologists read the full volume across planes with sequence and contrast context; a slice ambiguous alone is often unambiguous in the stack.
- **No localisation.** The model returns a class, not a segmentation or bounding box — it cannot say where the lesion is or how large.
- **Single-dataset evaluation.** MRI appearance varies with scanner vendor, field strength and protocol. Performance on another site's scans is unmeasured and should not be assumed.
- **Uncalibrated confidence.** The reported number is a raw softmax output, and deep networks are typically overconfident. A stated 97% should not be read as a well-calibrated probability without temperature scaling.
- **Not a medical device.** Unvalidated for clinical use, with no regulatory approval.

8. FUTURE WORK

- **Attack the dominant error mode directly** — class-weighted or focal loss biased toward the glioma/meningioma boundary, or a second-stage binary classifier invoked only when the top two probabilities are both in that pair.

- **Explainability.** Grad-CAM overlays would let a reader check whether the network attended to the lesion or to an irrelevant artefact — a prerequisite for any clinical conversation.
- **Volume-level inference** — aggregate across the slices of a study rather than classifying one.
- **External validation** on an independently acquired dataset, to measure the real generalisation gap.
- **Confidence calibration** via temperature scaling.
- **Larger backbones** — ResNet50, EfficientNet, a ViT — benchmarked to see whether capacity or data is the binding constraint.

9. CONCLUSION

A ResNet18 fine-tuned in two phases classifies brain MRI slices into four categories at **97.00% test accuracy**, with a macro F1 of 0.9715 and perfect recall on the no-tumour class. The two-phase schedule is what makes this work on 5,000 images: freezing the backbone while the new head stabilises, then adapting only the deepest block at a reduced learning rate, lifted validation accuracy from an 84.2% ceiling to 98.0%.

The residual errors are concentrated rather than diffuse — 63% of them sit on the glioma/meningioma boundary, the same distinction that is hardest for human readers on a single slice. That concentration is itself useful: it points at a specific, addressable target for the next iteration rather than a general need for more capacity.

Disclaimer. Research and educational use only. This model is not a medical device and is not validated for clinical diagnosis. Always consult a qualified medical professional.